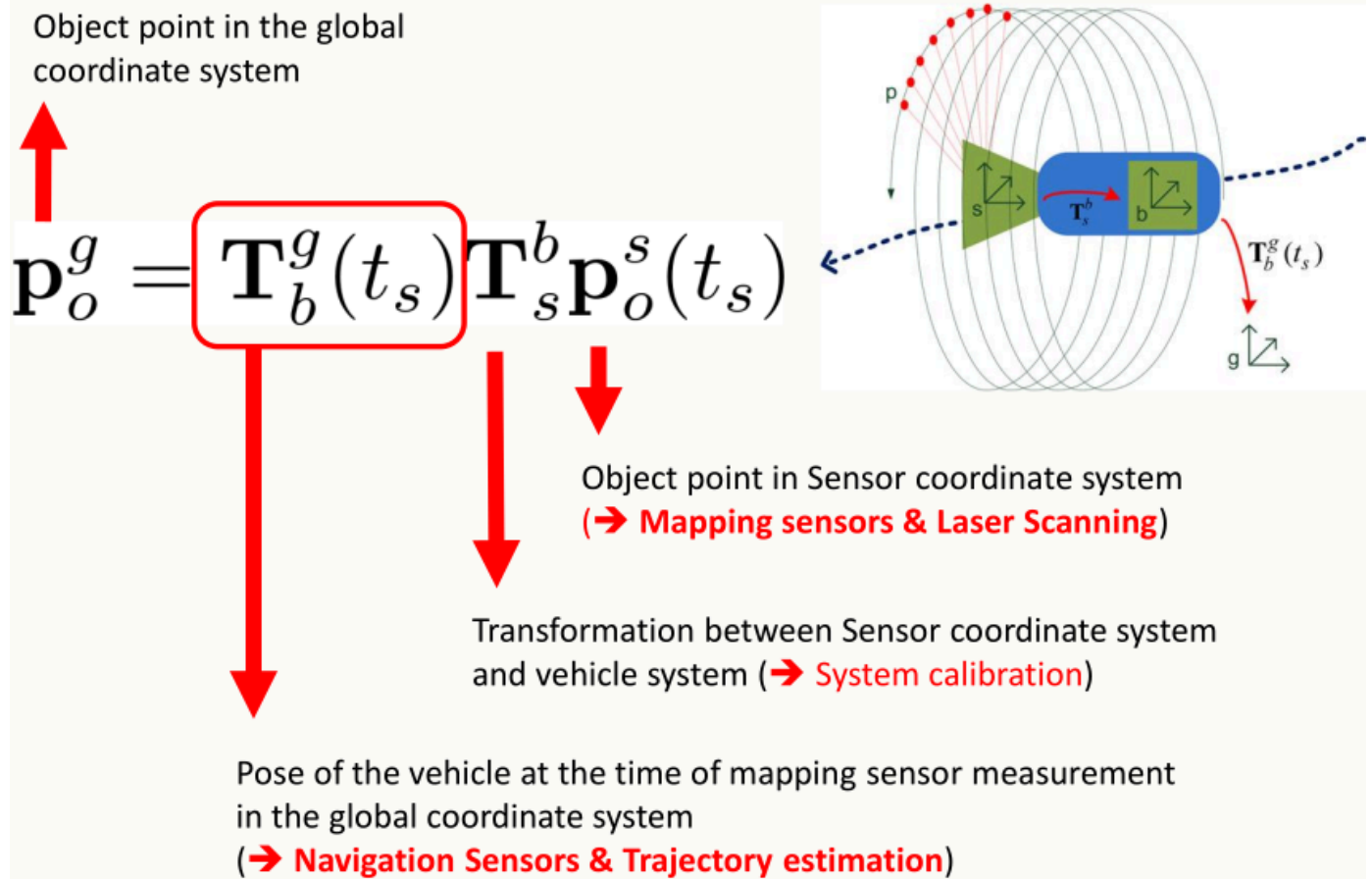
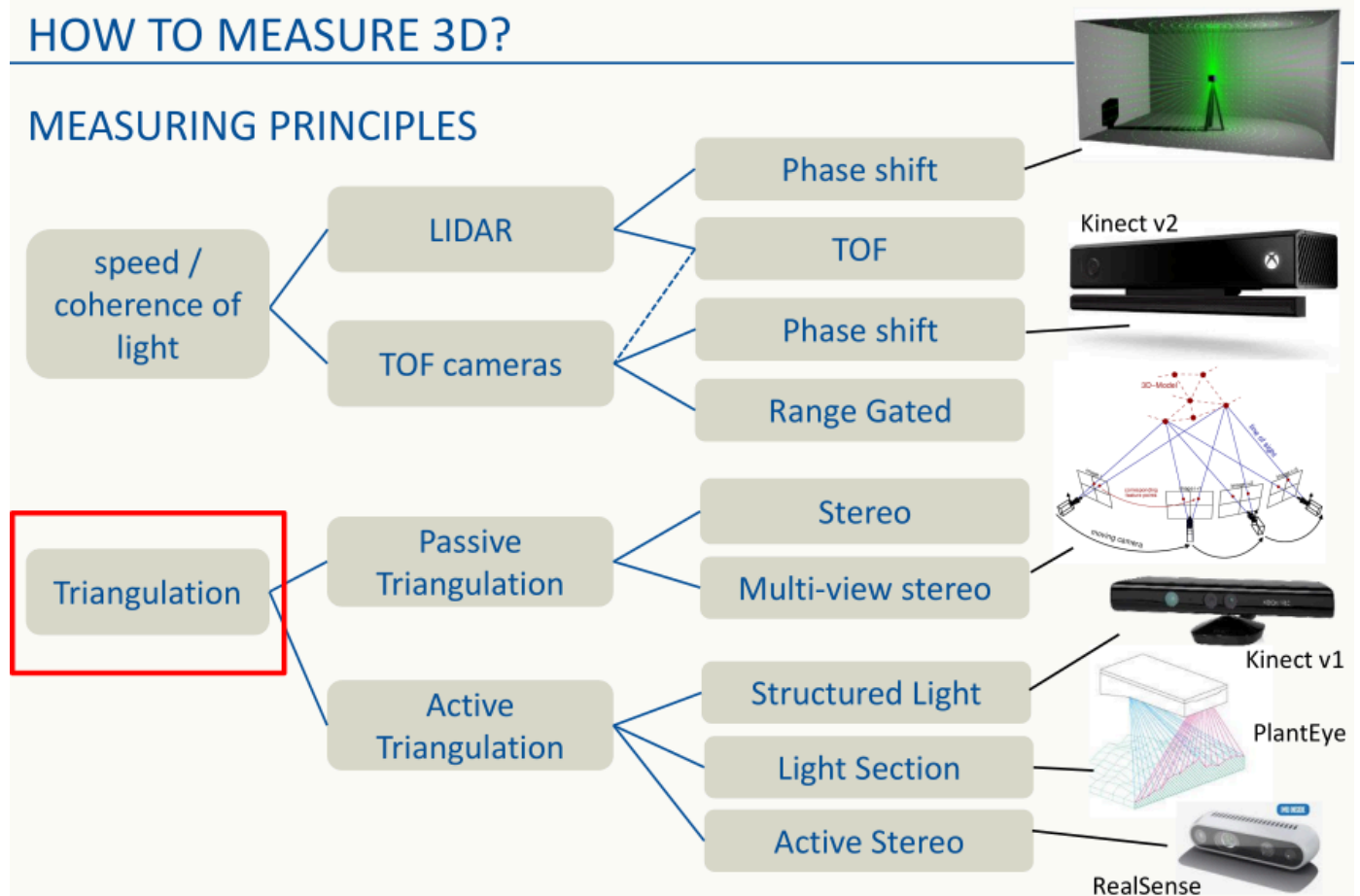


L1: 3D Sensors



HOW TO MEASURE 3D?

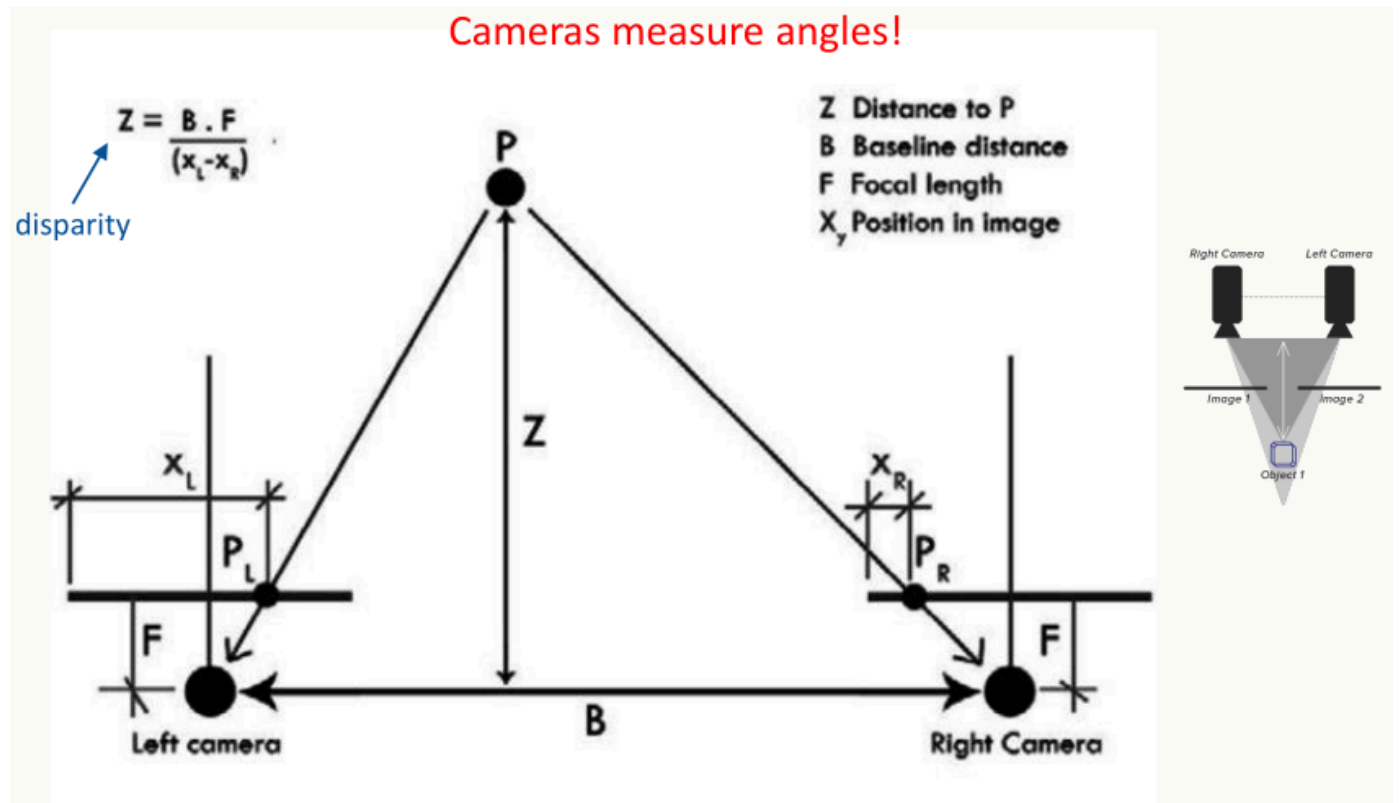
MEASURING PRINCIPLES



Triangulation

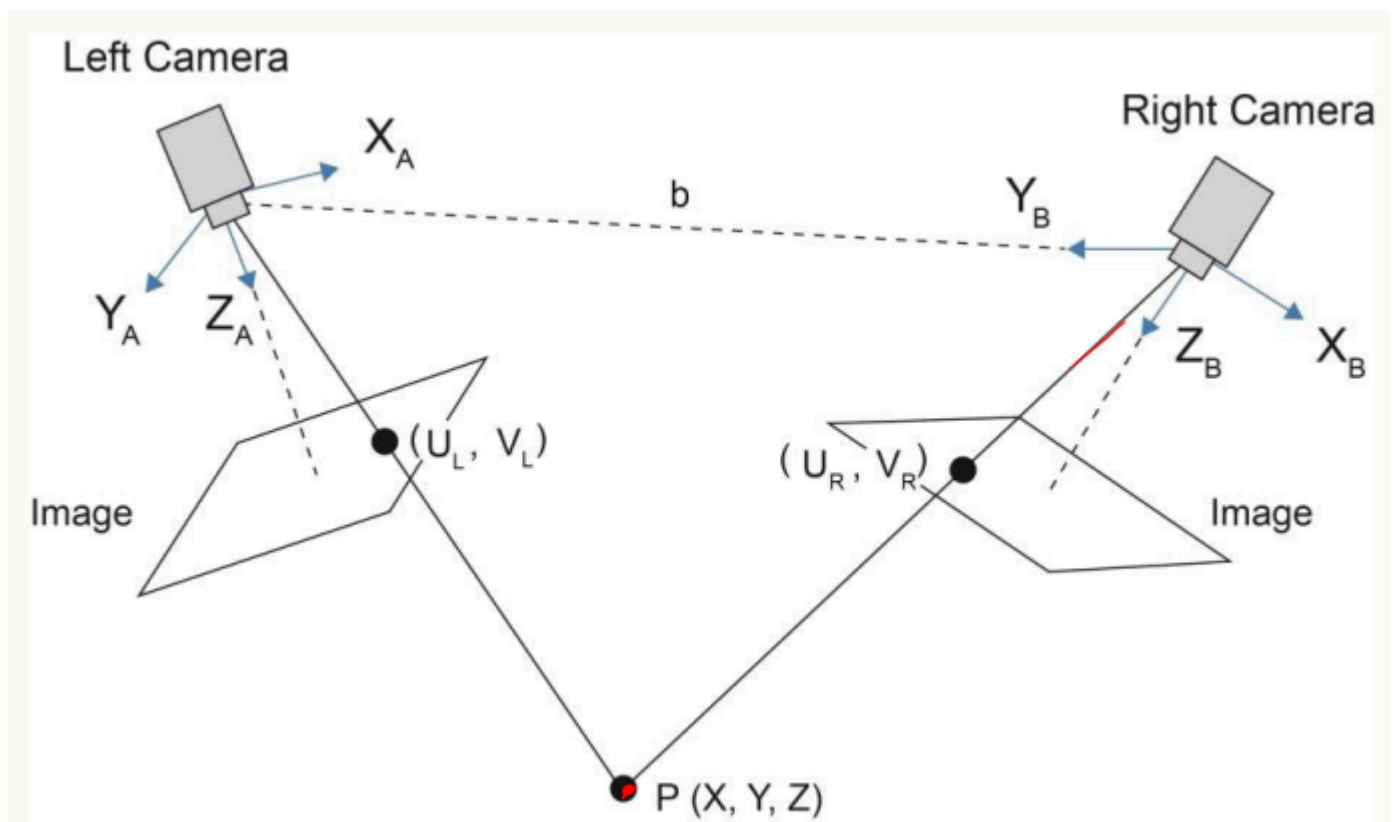
- Knowing 3 parameters of triangle lets us calculate other ones

Passive Triangulation



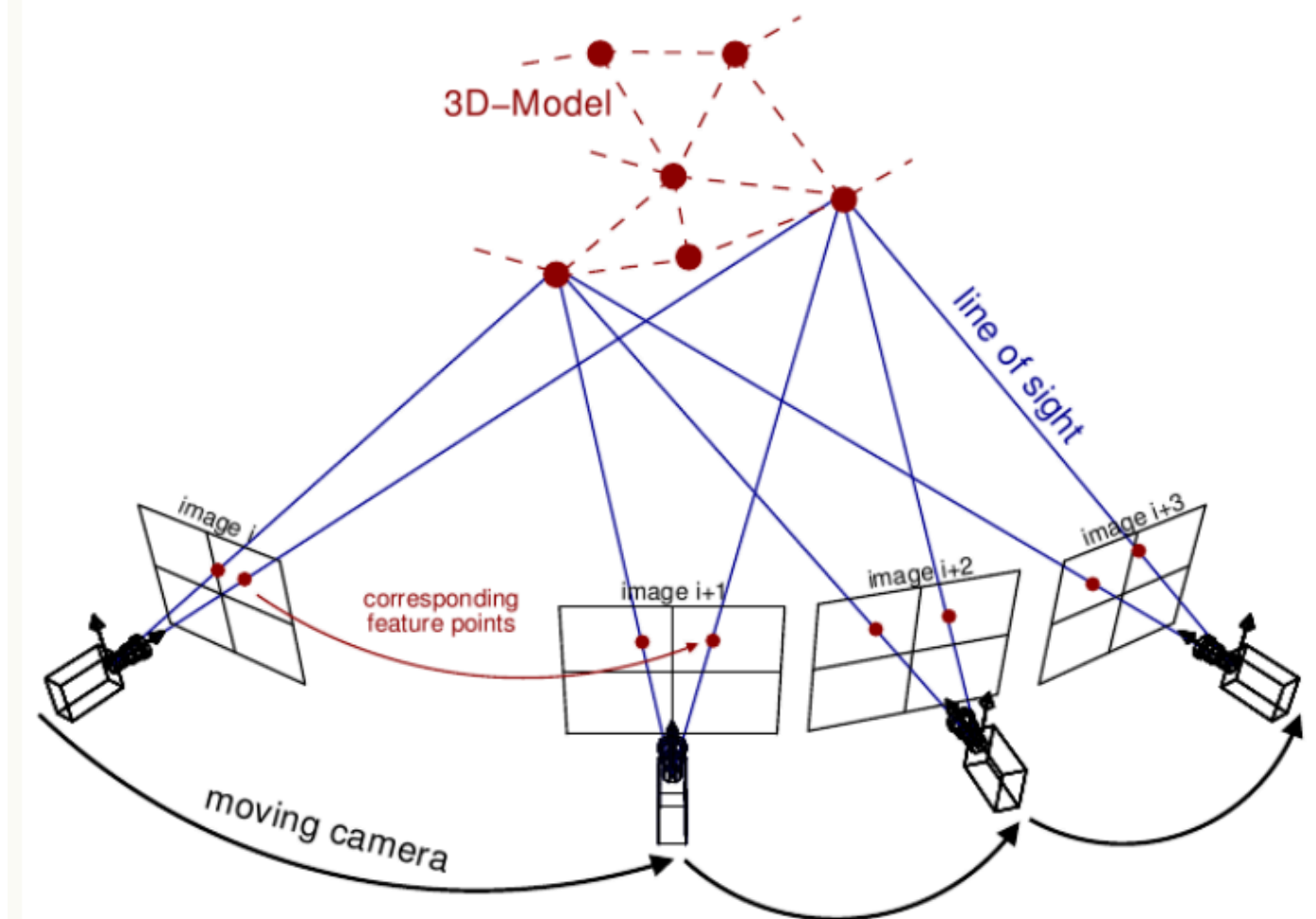
Stereo Vision

- Seeing point in 2 cameras And knowing relative camera pose
 - calculate distance of the point to cameras



- Obtain point cloud from depth image

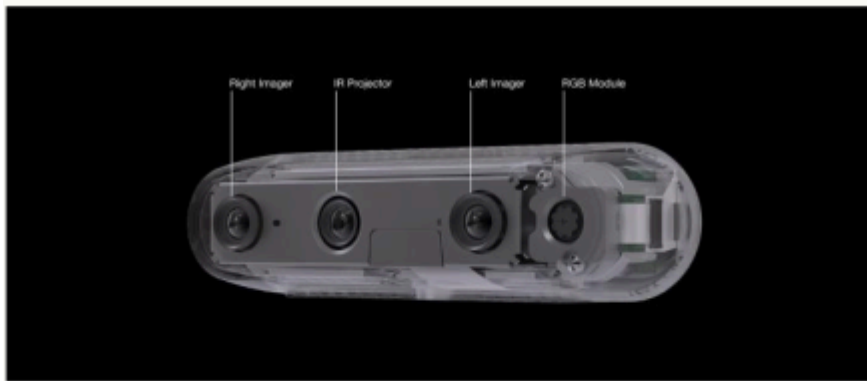
Multi View Stereo



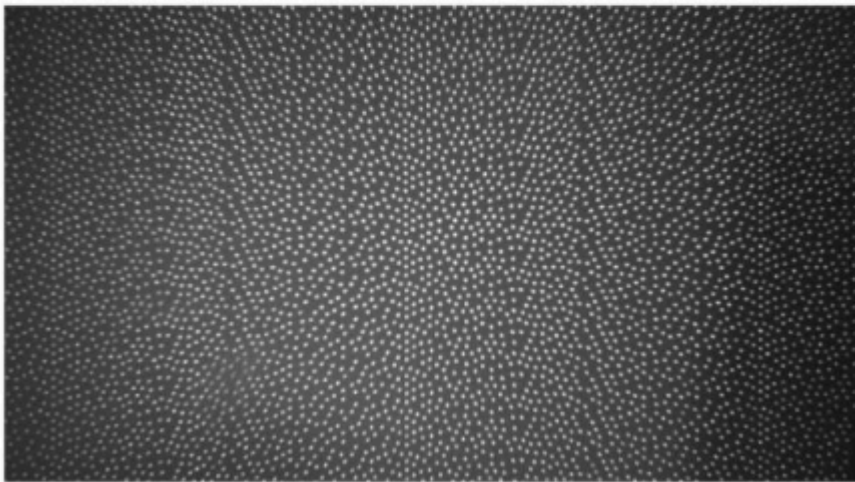
Active Triangulation

Active Stereo

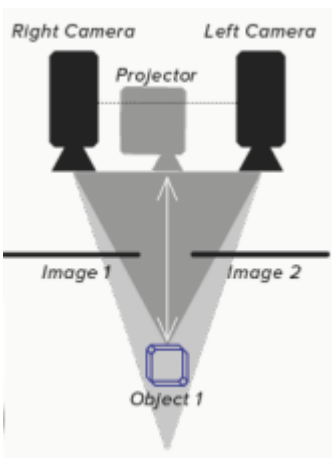
- Additional infrared dot pattern provides extra texture to the scene
- Helps for unstructured object (e.g. walls)



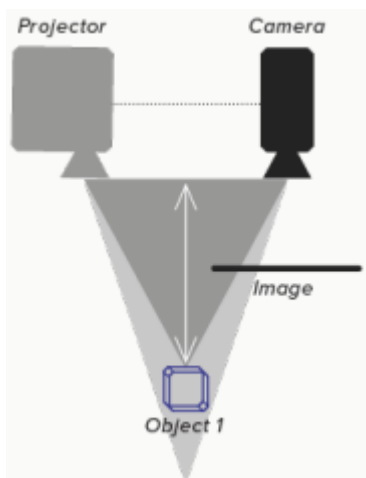
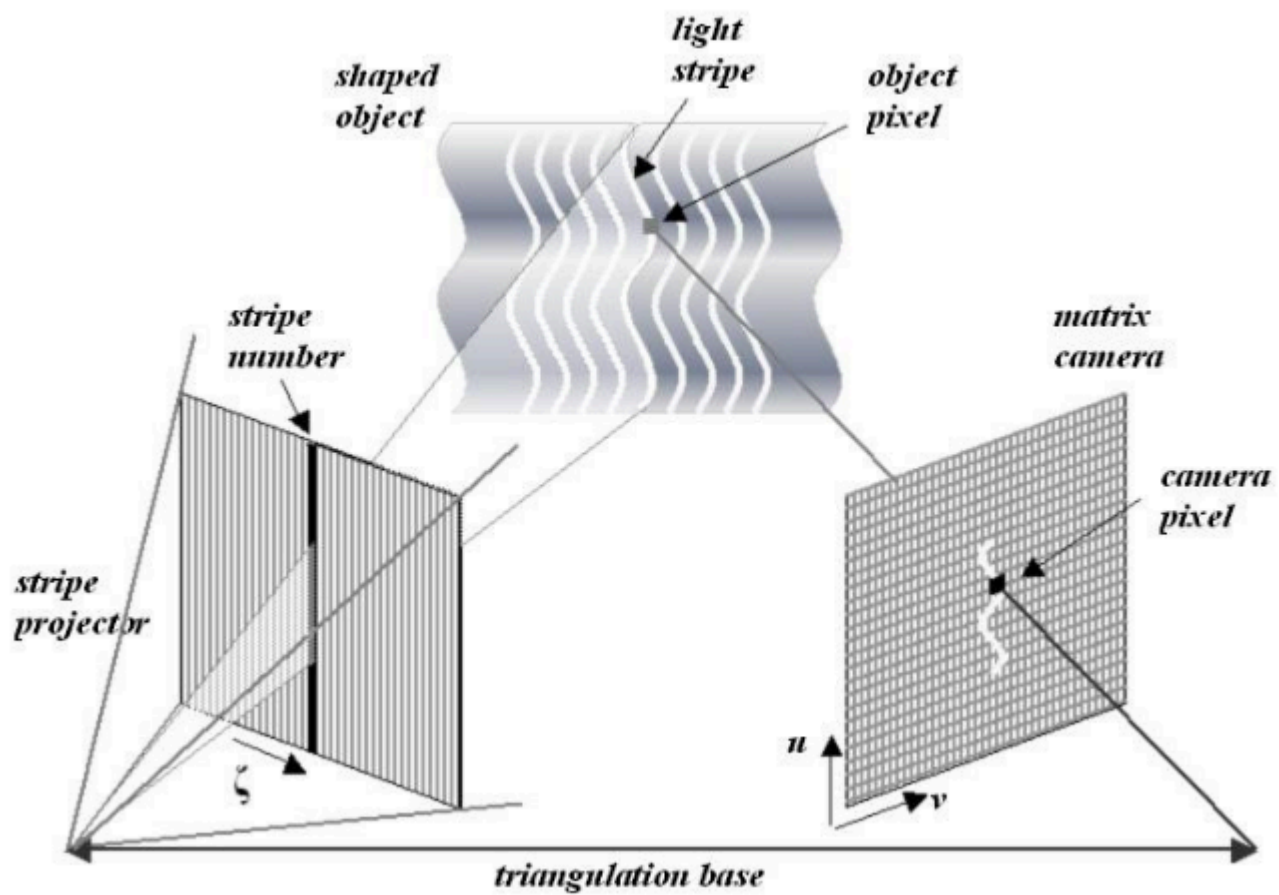
D435



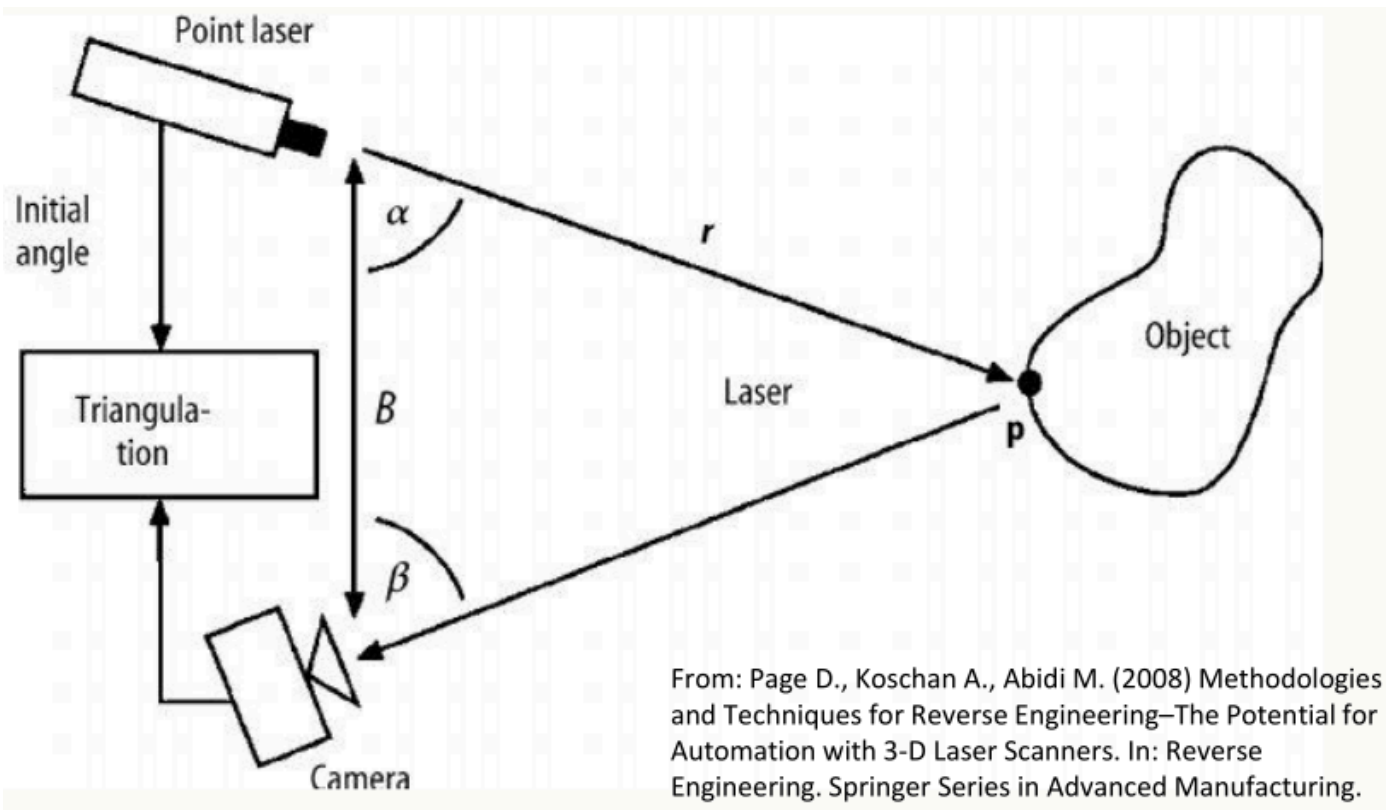
91x65 deg



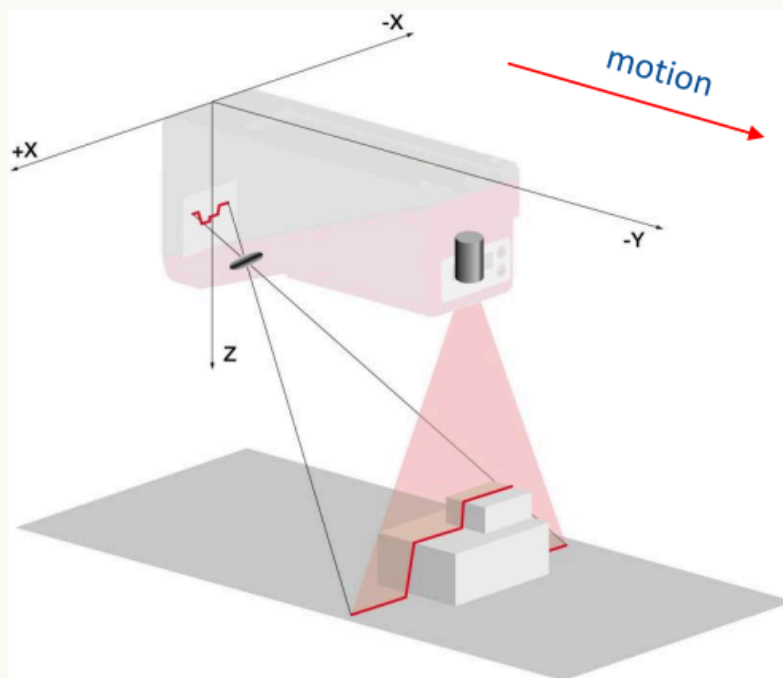
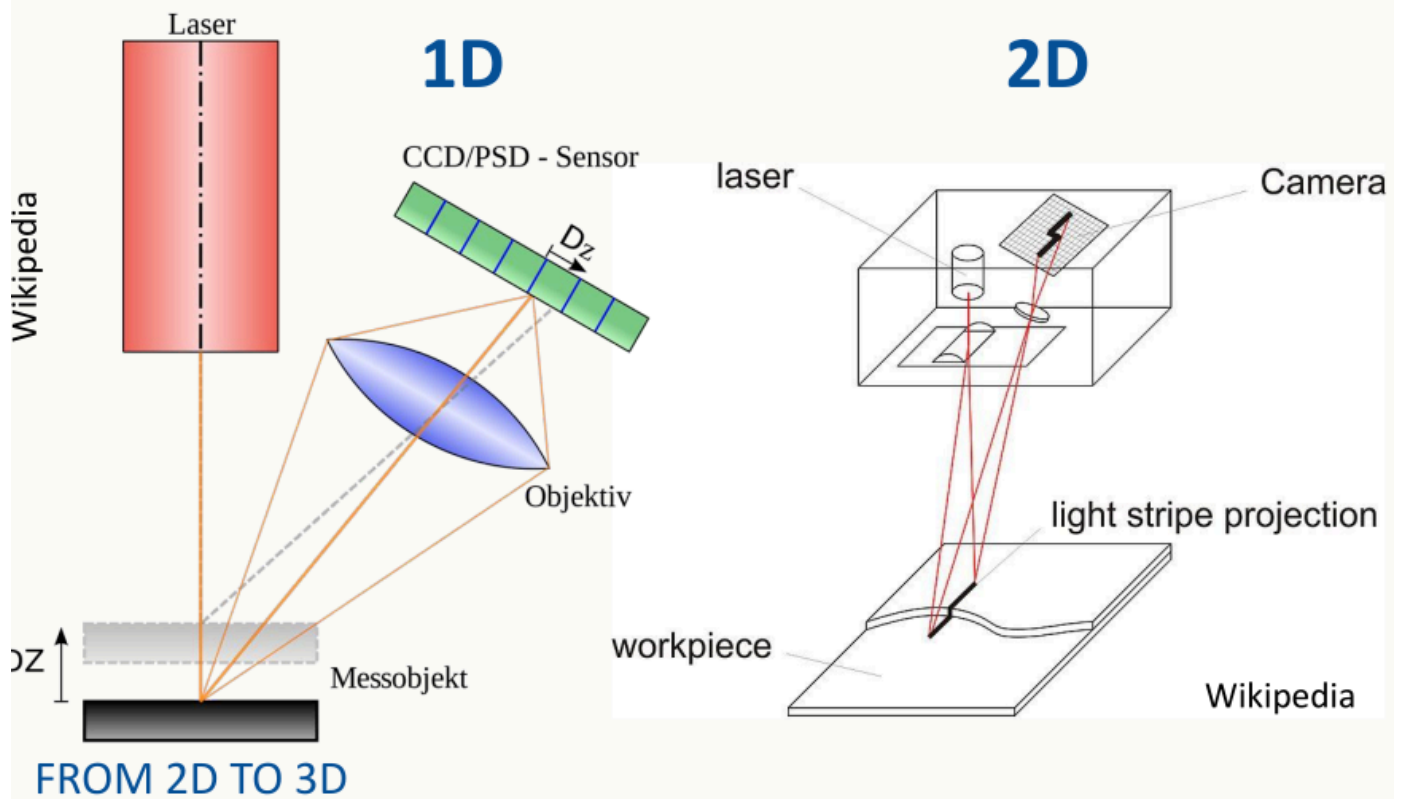
Structured Light Scanning



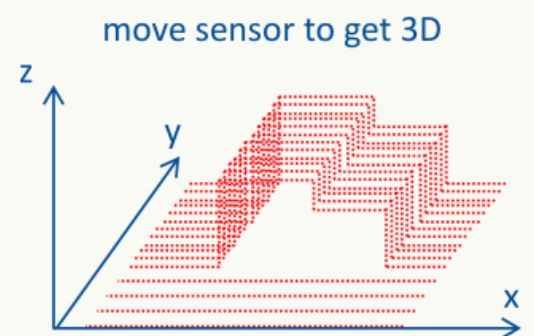
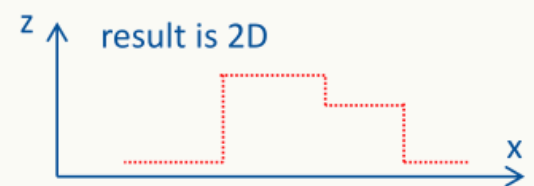
• Laser Based /Light Section



LASER BASED/LIGHT SECTION

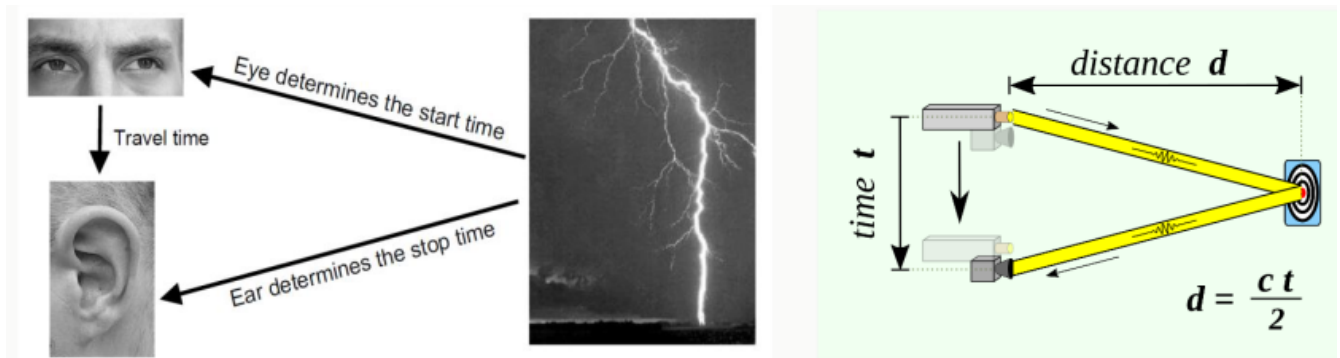


Leuze electronic GmbH + Co. KG

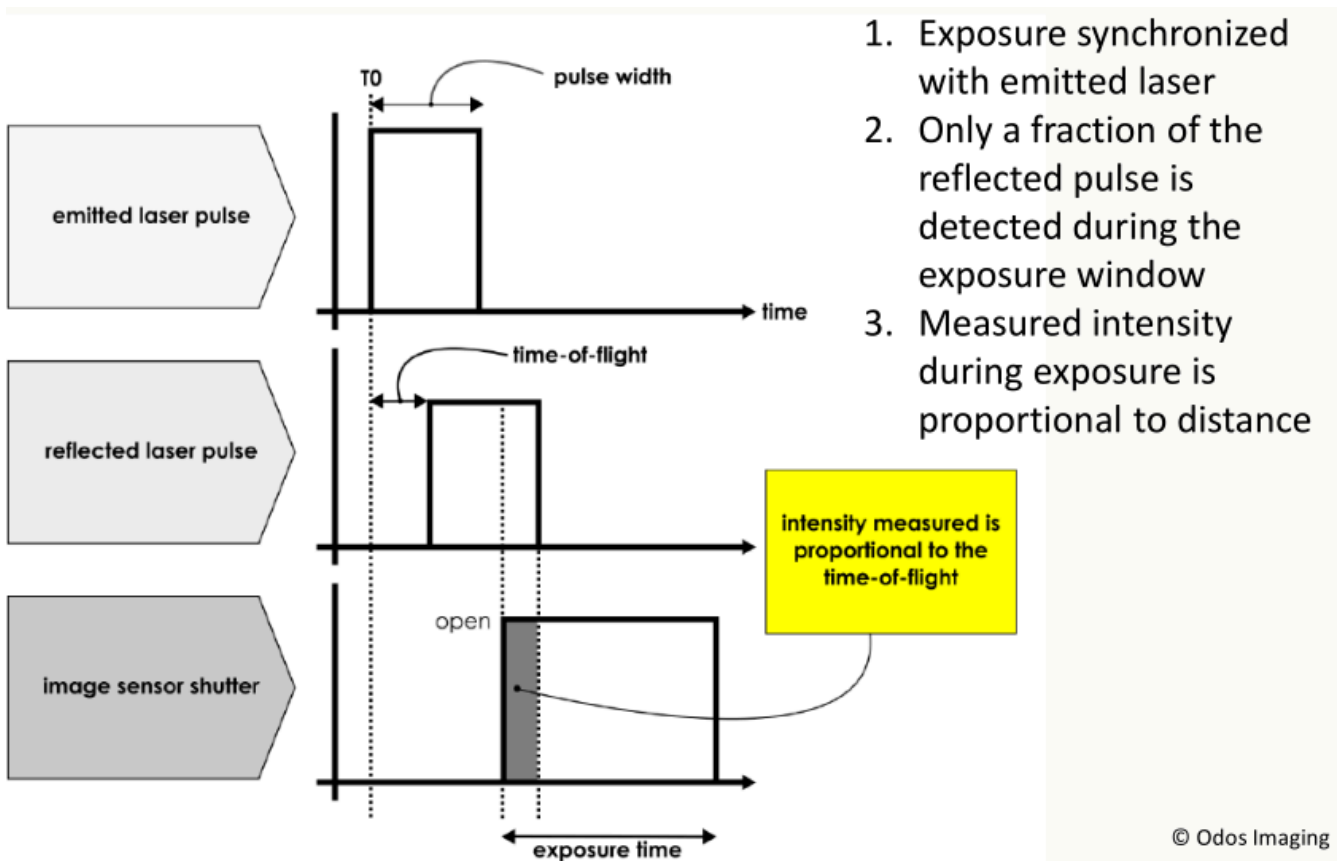


Speed/Coherence of Light

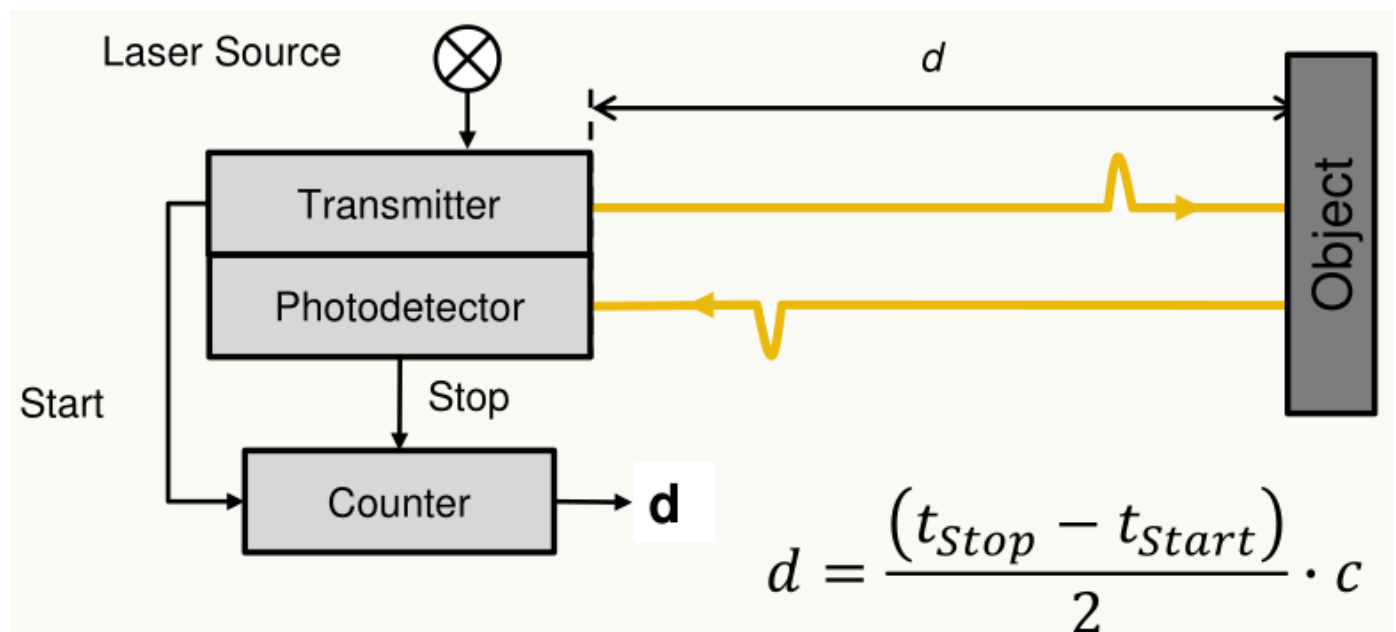
- Relation between distance and travel time between sensor and object



TOF Cameras

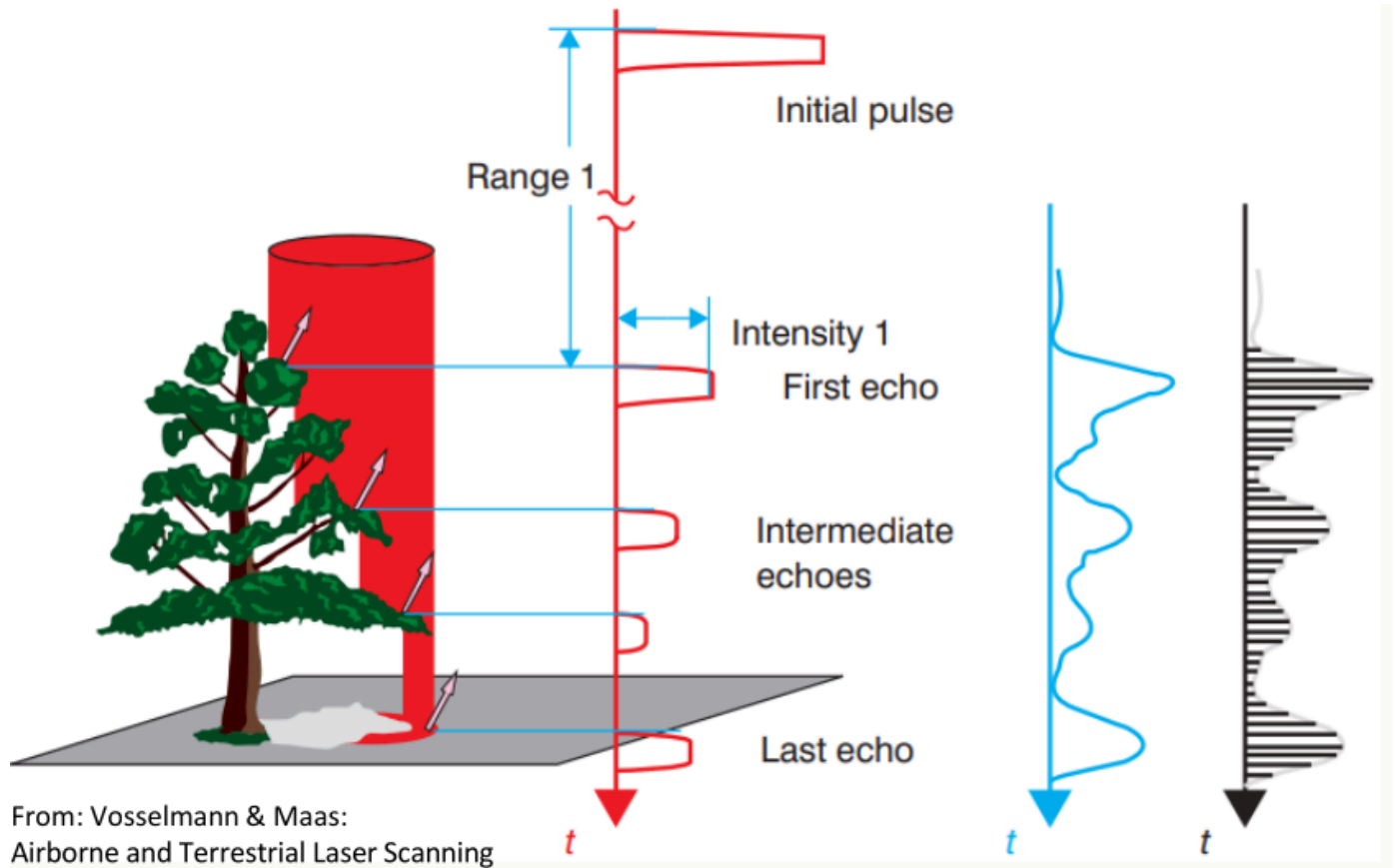


LIDAR



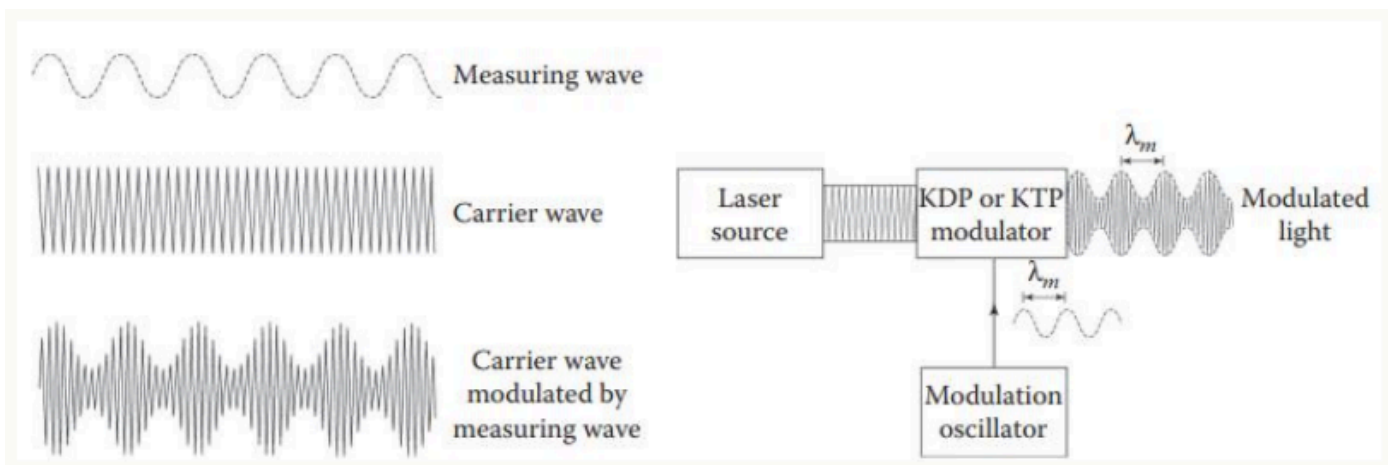
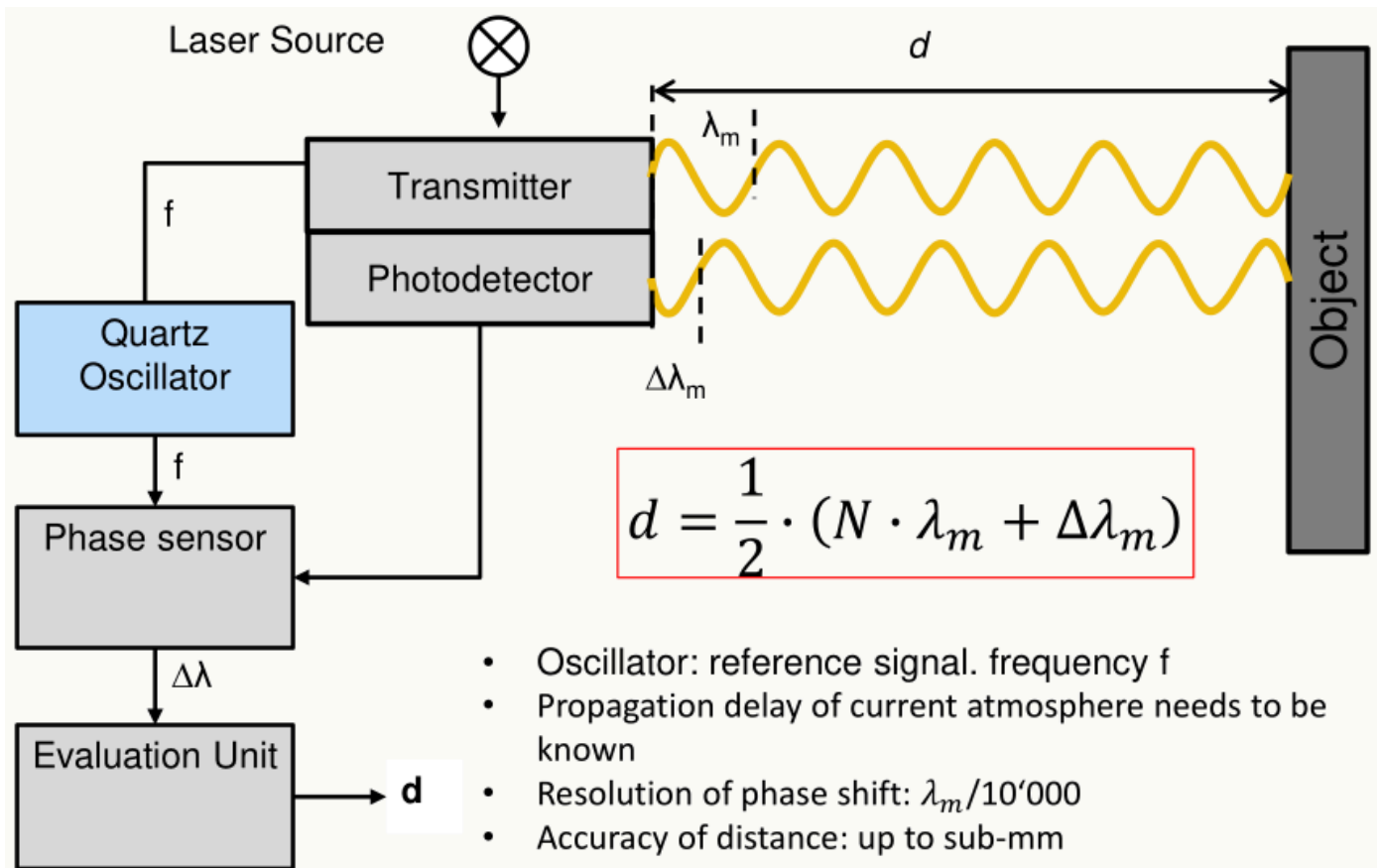
- c : Speed of light
- Atmospheric propagation delay known
- Resolution of clock: picoseconds
- Accuracy of distance: mm

TOF + Waveform-Digitising (WFD)



- Scanners provide
 - Full sampled signal (some Riegl scanner)
 - Multiple returns plus additional information derived from the signal (some Riegl scanners)
 - Distance of highest peak
 - Multiple returns (e.g. 2 or 4)

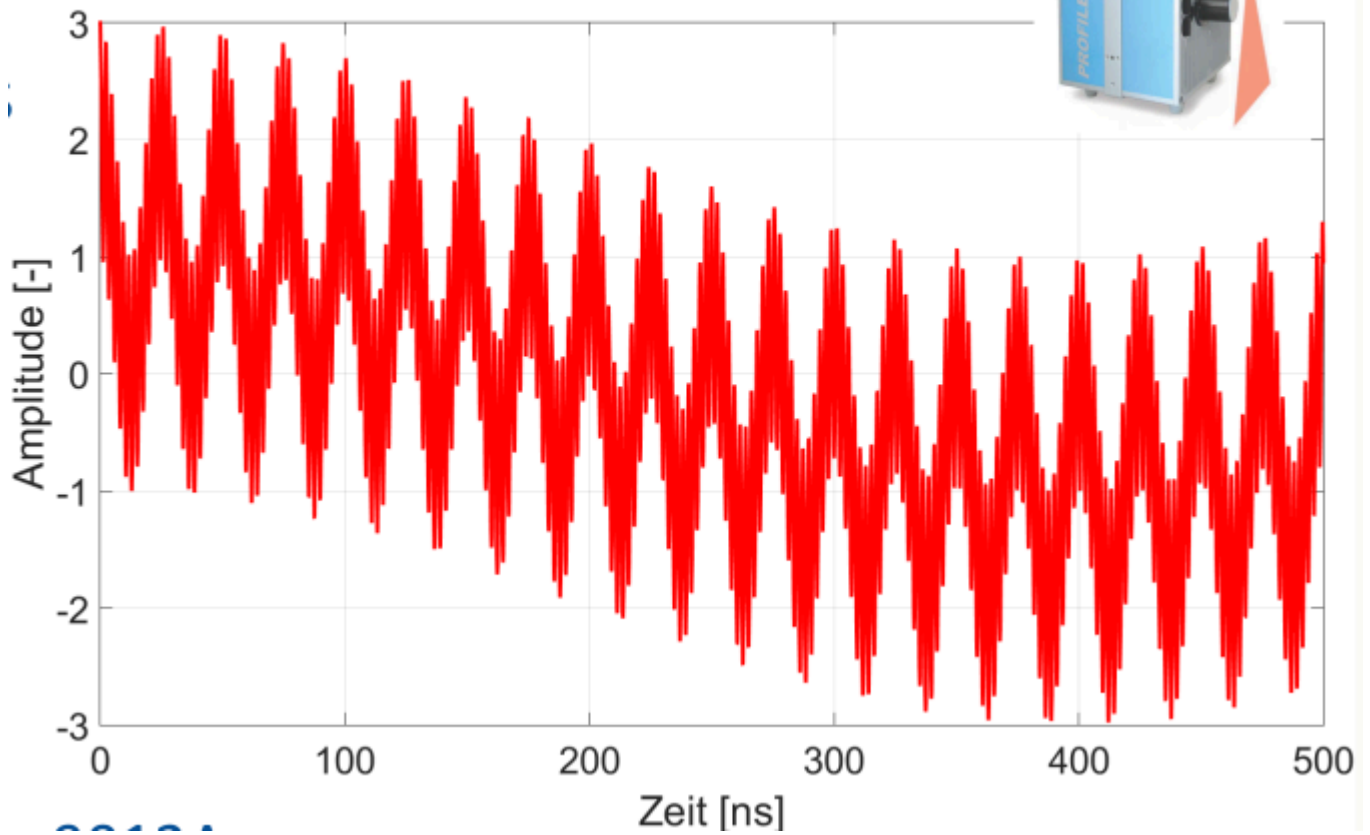
Phase Shift



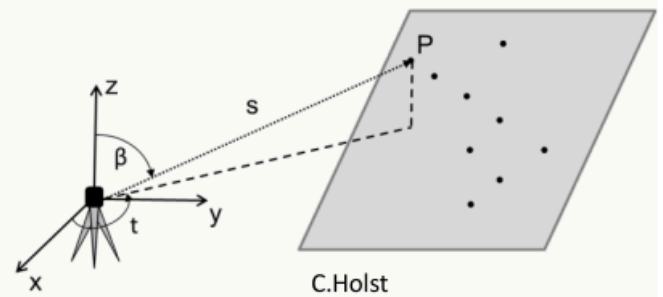
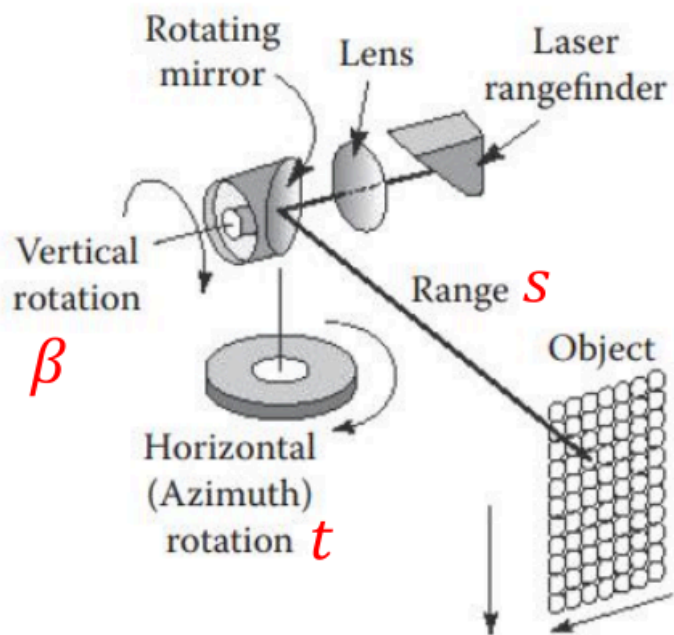
- Use of multiple frequencies allow resolution of ambiguities
- Minimal freq. determines resolution
- Maximal freq. determines max range

Phase shift: Resolving ambiguities

T



SAMPLING THE ENVIRONMENT

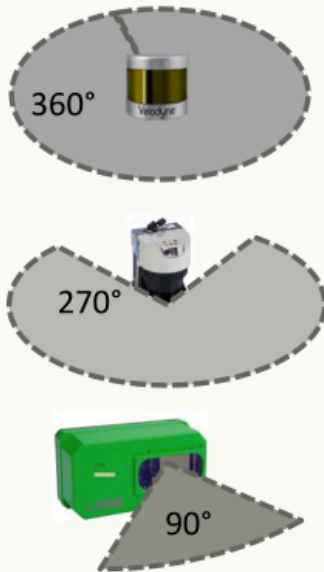


$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} s \cdot \sin \beta \cdot \cos t \\ s \cdot \sin \beta \cdot \sin t \\ s \cdot \cos \beta \end{bmatrix}$$

Scanner Types

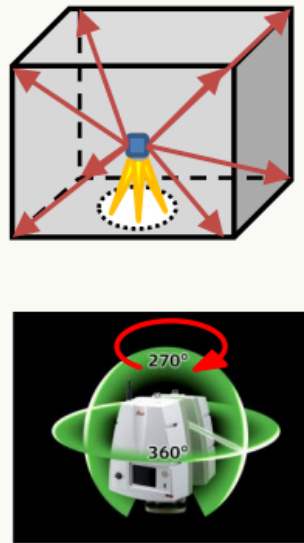
- Mixtures are possible

PROFILE SCANNER



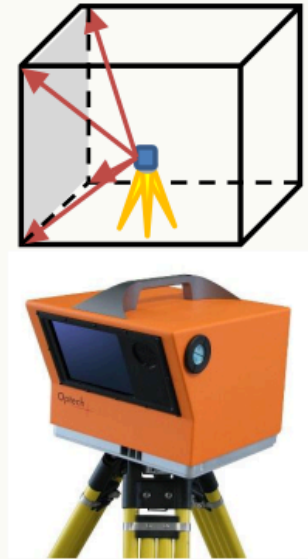
Needs motion to cover an area

PANORAMA SCANNER

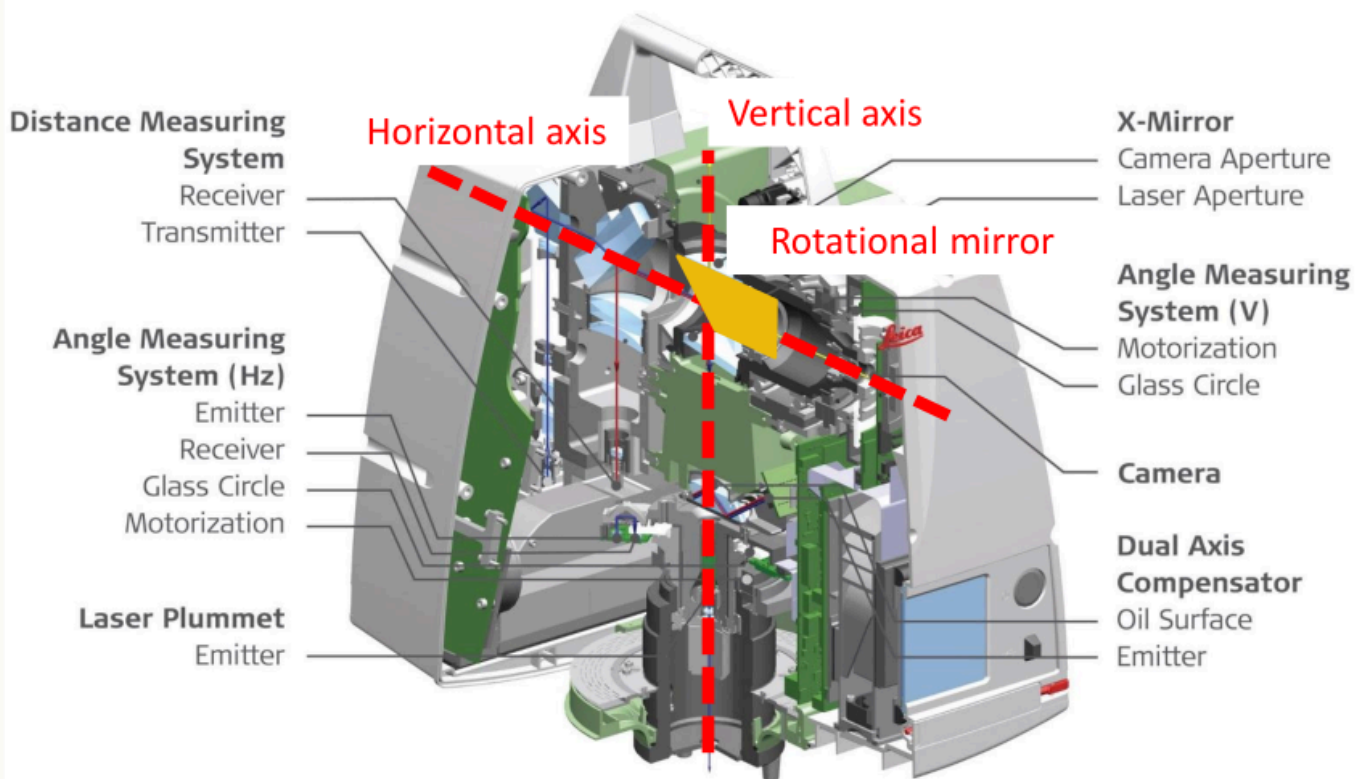


360° Scan

CAMERA SCANNER



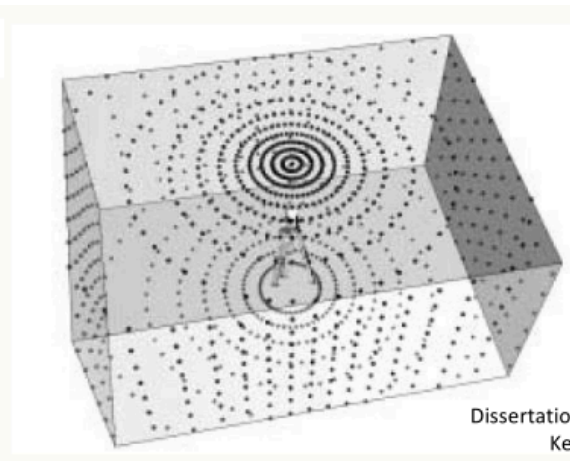
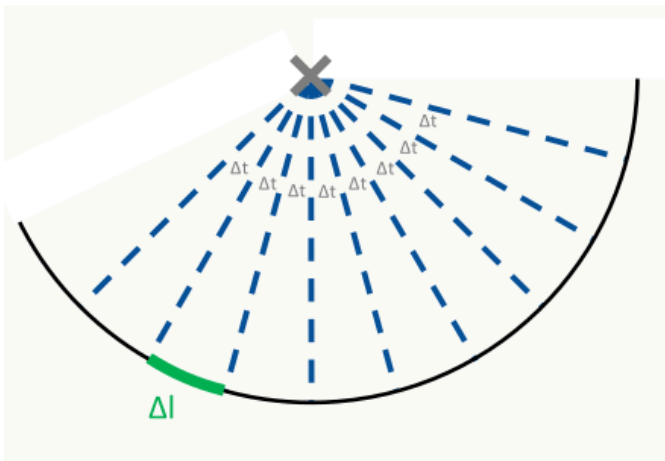
Specific Field of view



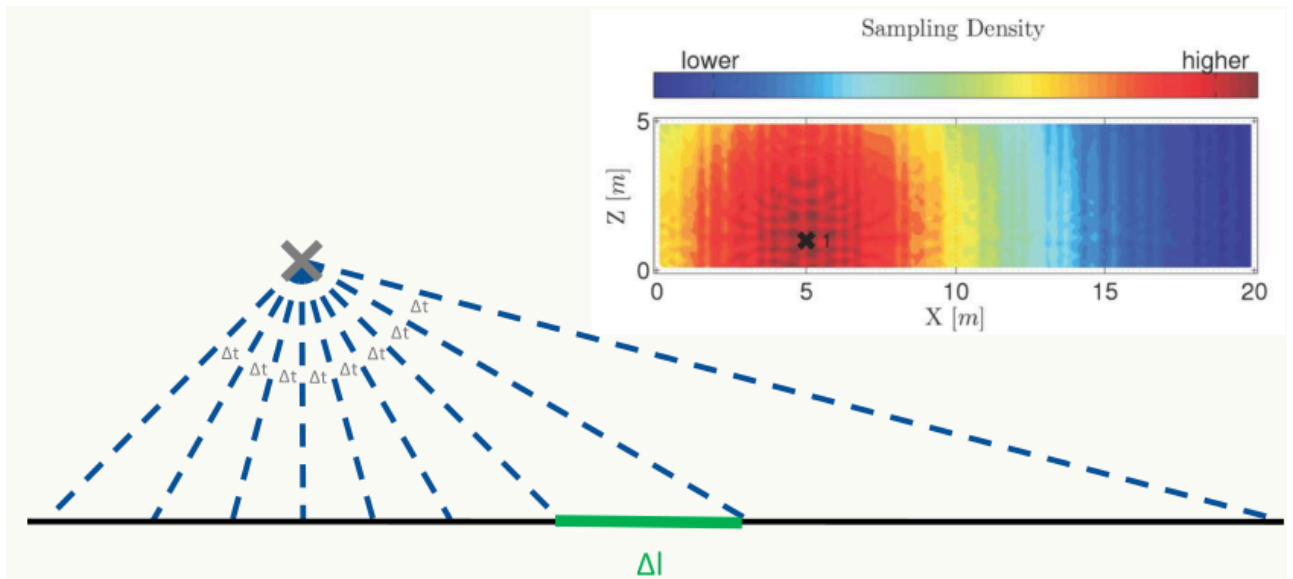
Leica

Sampling of Environment

- Sampling spherically regular -> $\Delta t = \text{constant}$, $\Delta \beta = \text{constant}$



-
- Sampling irregular in Cartesian space
 - -> Strong variation of sampling density dependent on measuring geometry



-
- 3D Point cloud
- Unstructured collection of points with x,y,z coordinates in some coordinate system
 - Plus: optional additional parameters, such as color (r,g,b)